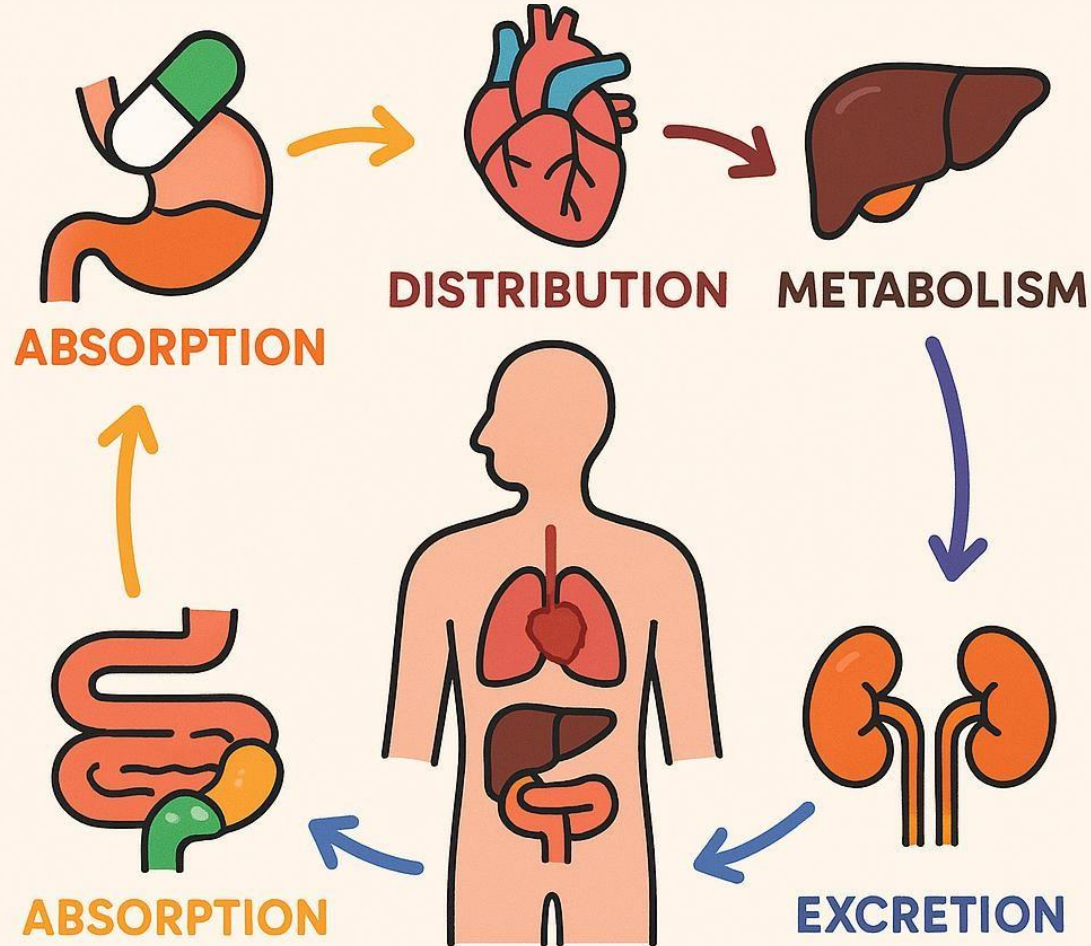




Pharmacokinetics

Dr Tiffany Clementson

Objectives

- SOM.MK.1.BPM1.1.FTM.4.PHAR.0013 Describe the mechanisms of drug permeation through cell membranes.
- SOM.MK.1.BPM1.1.FTM.4.PHAR.0014 Define drug absorption.
- SOM.MK.1.BPM1.1.FTM.4.PHAR.0015 Describe the main routes of drug administration. Explain the first-pass effect.
- SOM.MK.1.BPM1.1.FTM.4.PHAR.0016 Describe the factors that influence drug absorption.
- SOM.MK.1.BPM1.1.FTM.4.PHAR.0017 Explain the effect of pH on drug absorption. Describe and use the Henderson-Hasselbalch equation. Explain ion trapping.
- SOM.MK.1.BPM1.1.FTM.4.PHAR.0018 Explain why the intestine is the major site of drug absorption.
- SOM.MK.1.BPM1.1.FTM.4.PHAR.0019 Describe how P-glycoprotein affects drug absorption.
- SOM.MK.1.BPM1.1.FTM.4.PHAR.0020 Define bioavailability. Describe how bioavailability is determined.
- SOM.MK.1.BPM1.1.FTM.4.PHAR.0021 Define drug distribution. Describe the various factors that affect drug distribution.
- SOM.MK.1.BPM1.1.FTM.4.PHAR.0022 List the major mechanisms responsible for drug elimination.
- SOM.MK.1.BPM1.1.FTM.4.PHAR.0023 Describe how metabolism affects drug structure and function. List the main organs involved in drug metabolism.
- SOM.MK.1.BPM1.1.FTM.4.PHAR.0024 Describe the role of the cytochrome P450 enzymes in drug metabolism. Explain the clinical relevance of enzyme induction and enzyme inhibition. List the main drugs that can act as inhibitors or inducers of cytochrome P450 isoenzymes.
- SOM.MK.1.BPM1.1.FTM.4.PHAR.0025 Describe the main mechanisms of drug excretion

PHARMACOKINETICS

- A drug should be able to reach its site of action after administration.
- In few situations a drug is directly applied to its target tissue.
 - Example: topical application of an anti-inflammatory agent to inflamed skin.



PHARMACOKINETICS

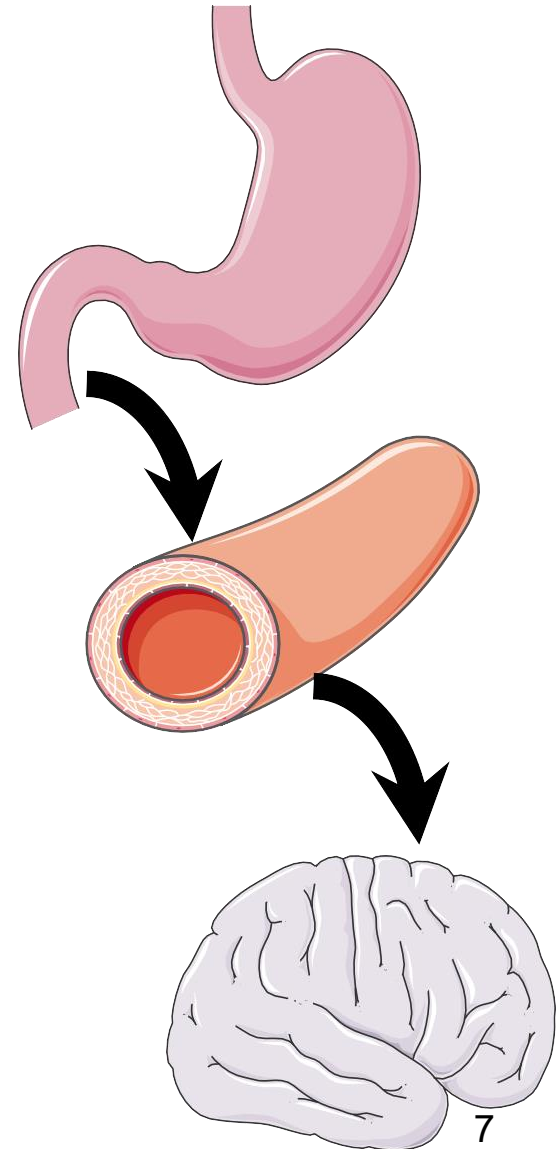
- Much more commonly, a drug is given into one body compartment, e.g. the gut, and must move to its site of action in another compartment, e.g. the brain.

PHARMACOKINETICS

- So, the drug must be **absorbed** into the blood from its site of administration and **distributed** to its **site of action**, **permeating through the various barriers** that separate these compartments.

PHARMACOKINETICS

- For a drug given **orally** to produce an effect in **the CNS** these barriers include:
 - The wall of the intestine
 - The walls of the capillaries that perfuse the gut
 - The blood-brain barrier



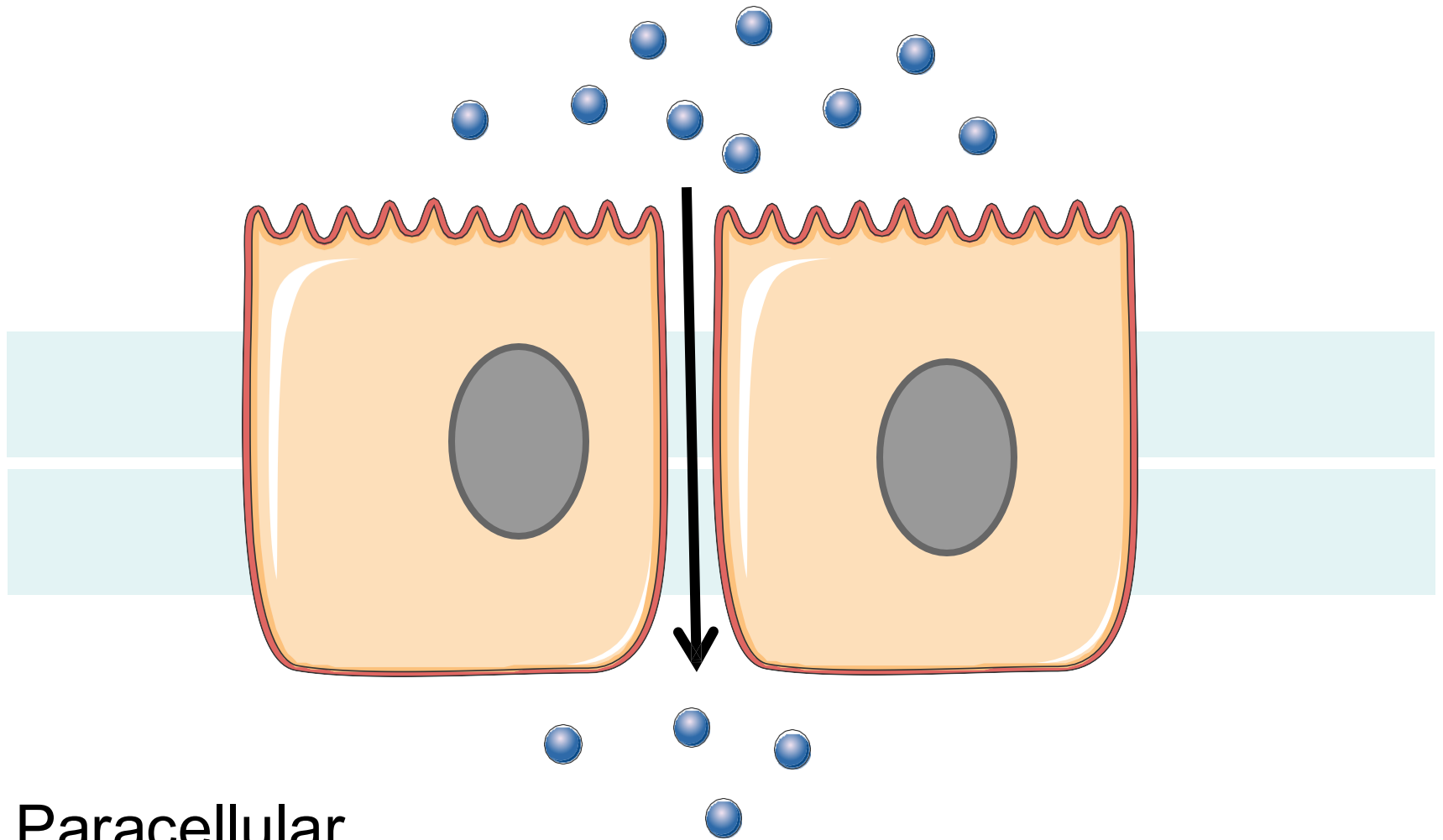
PHARMACOKINETICS

- Finally, after bringing about its effect, a drug should be **eliminated**.

MECHANISMS OF DRUG PERMEATION

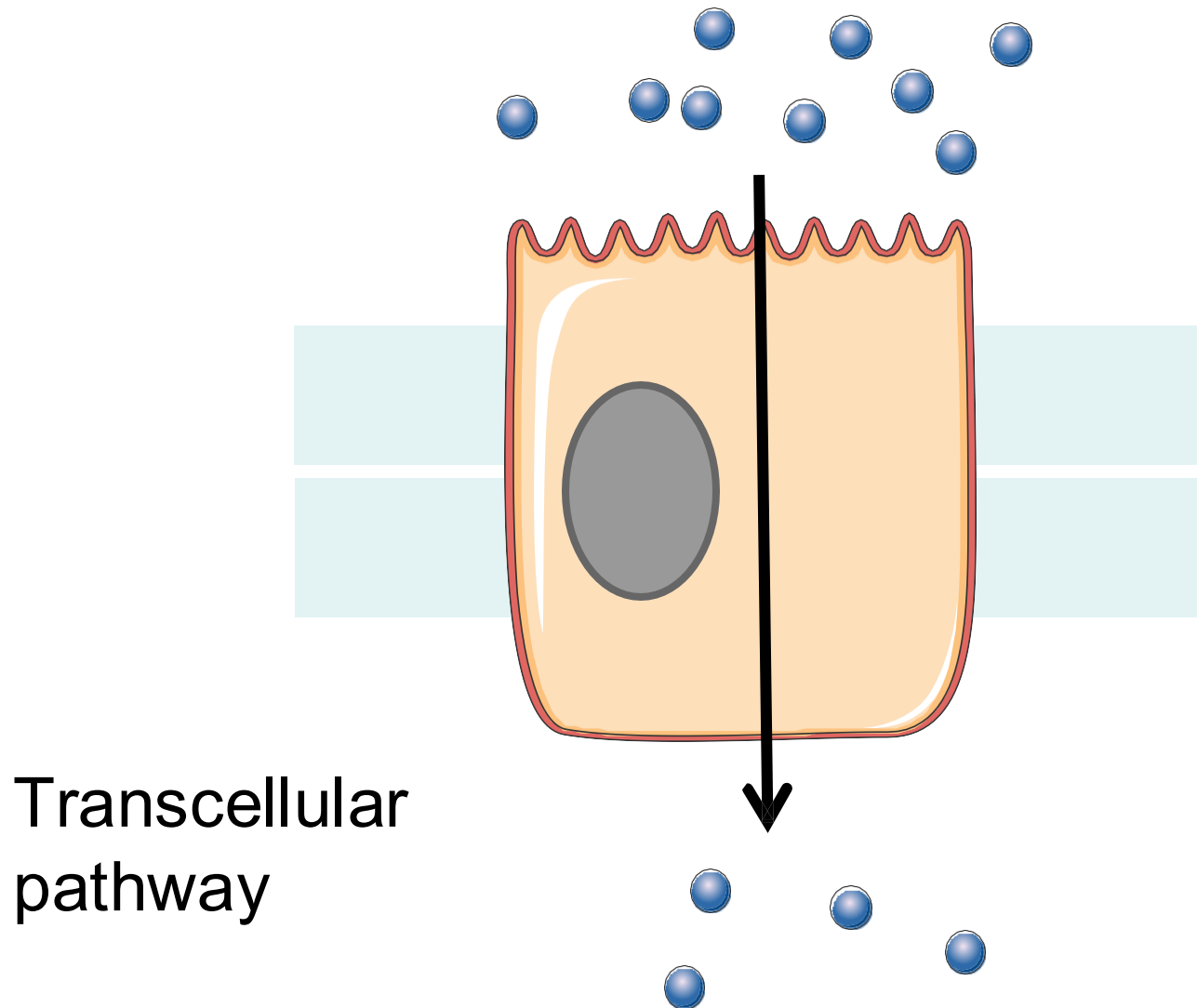
- **AQUEOUS DIFFUSION (PARACELLULAR PATHWAY)**
- **LIPID DIFFUSION (TRANSCELLULAR PATHWAY)**
- **SPECIAL CARRIERS**
- **ENDOCYTOSIS s EXOCYTOSIS (TRANSCYTOSIS)**

MECHANISMS OF DRUG PERMEATION

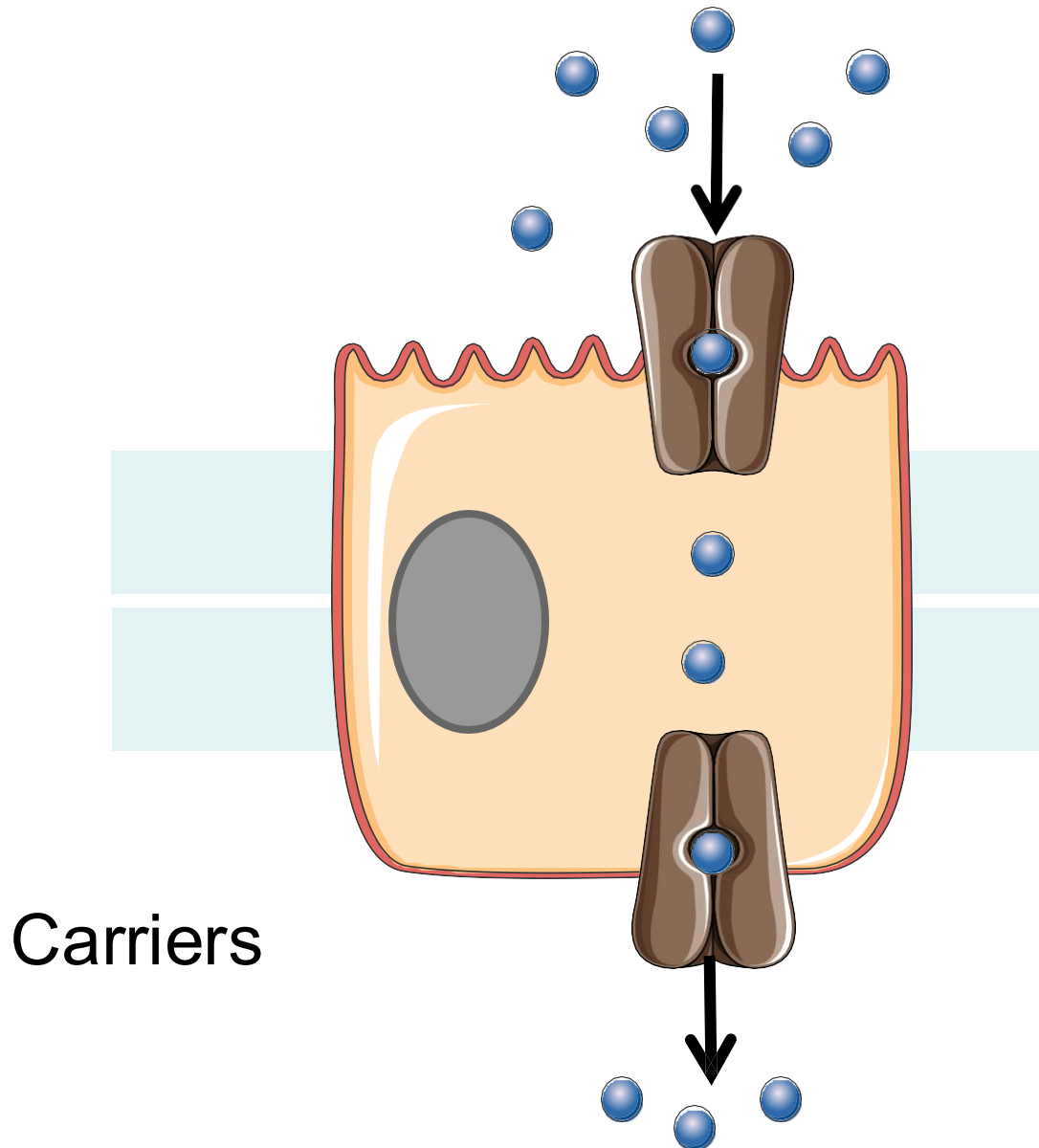


Paracellular
pathway

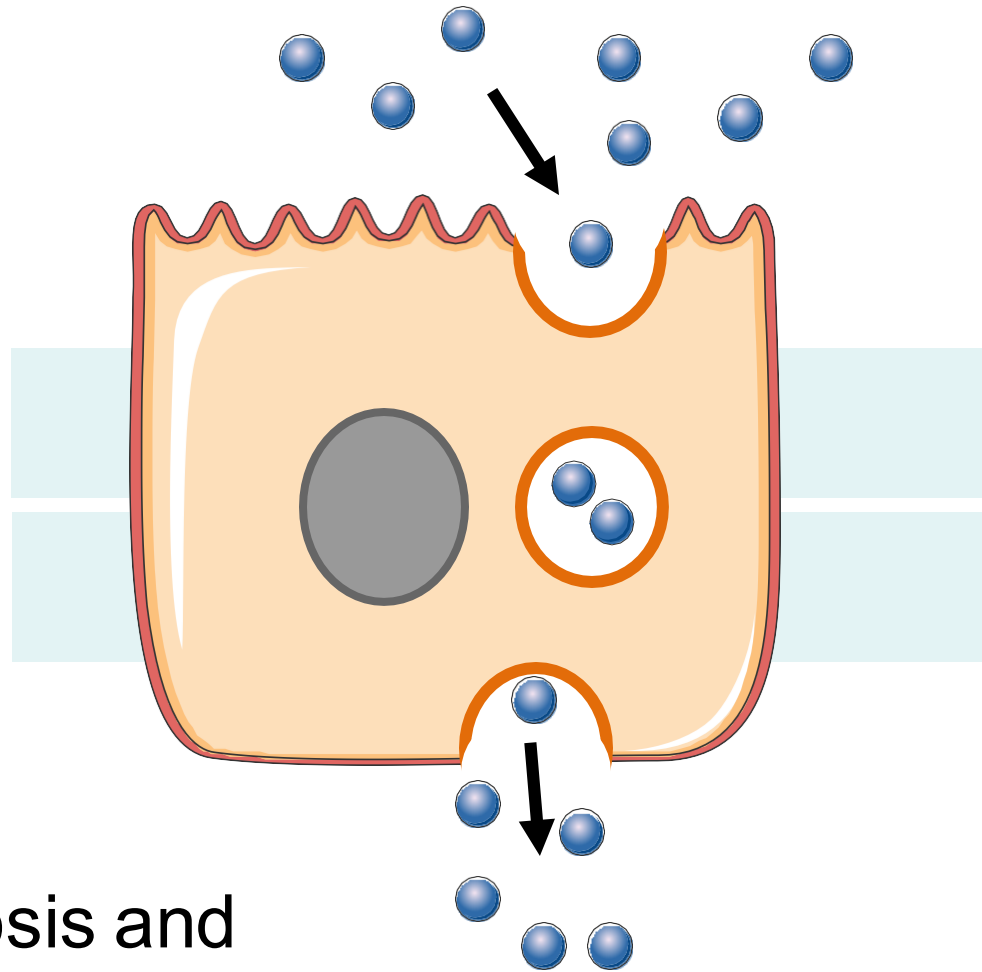
MECHANISMS OF DRUG PERMEATION



MECHANISMS OF DRUG PERMEATION

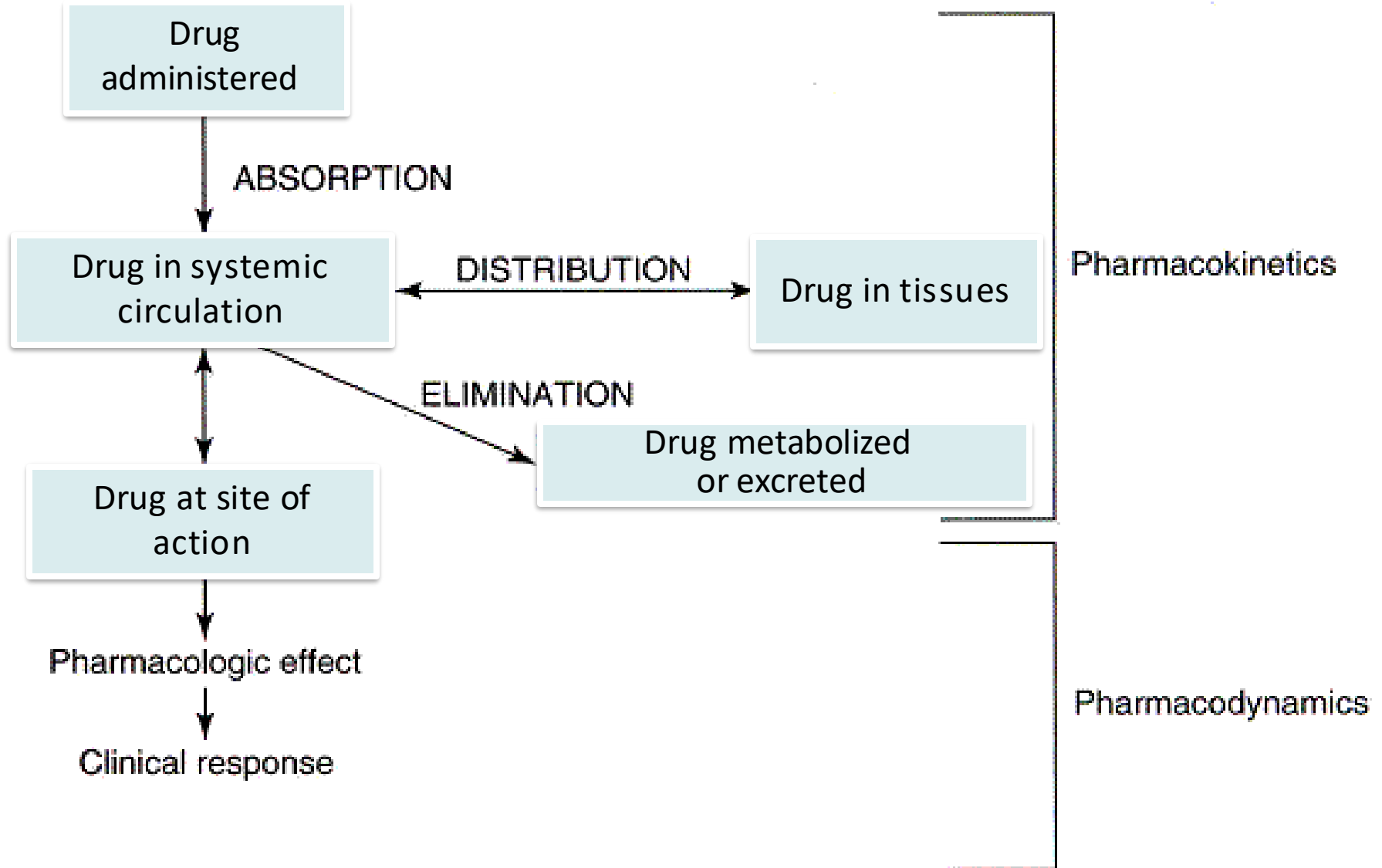


MECHANISMS OF DRUG PERMEATION



Endocytosis and
exocytosis

PHARMACOKINETICS & PHARMACODYNAMICS



ABSORPTION OF DRUGS

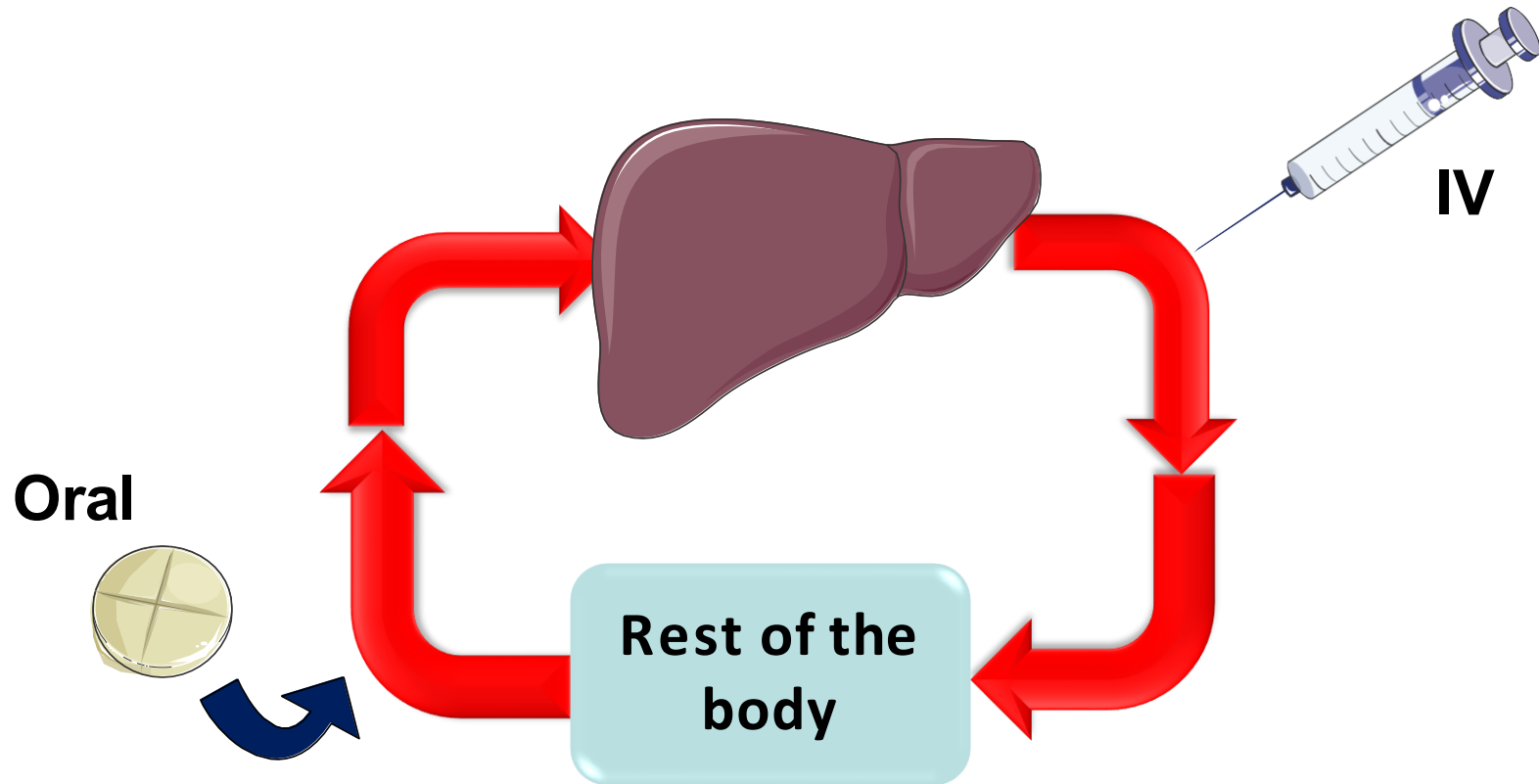
- Absorption = **transfer of a drug from its site of administration to the bloodstream.**
- The rate and efficiency of absorption depend on the route of administration.
- For IV delivery, **absorption is complete**: the total dose reaches systemic circulation.
- Drug delivery by other routes may result in only **partial absorption.**

ROUTES OF DRUG ADMINISTRATION

ENTERAL ROUTES

- **Oral:** maximum convenience, but absorption may be slower and less complete than with parenteral routes.
- Ingested drugs suffer **first-pass effect**: a fraction of the drug is **metabolized in the gut wall and the liver** before reaching systemic circulation.

THE FIRST-PASS EFFECT



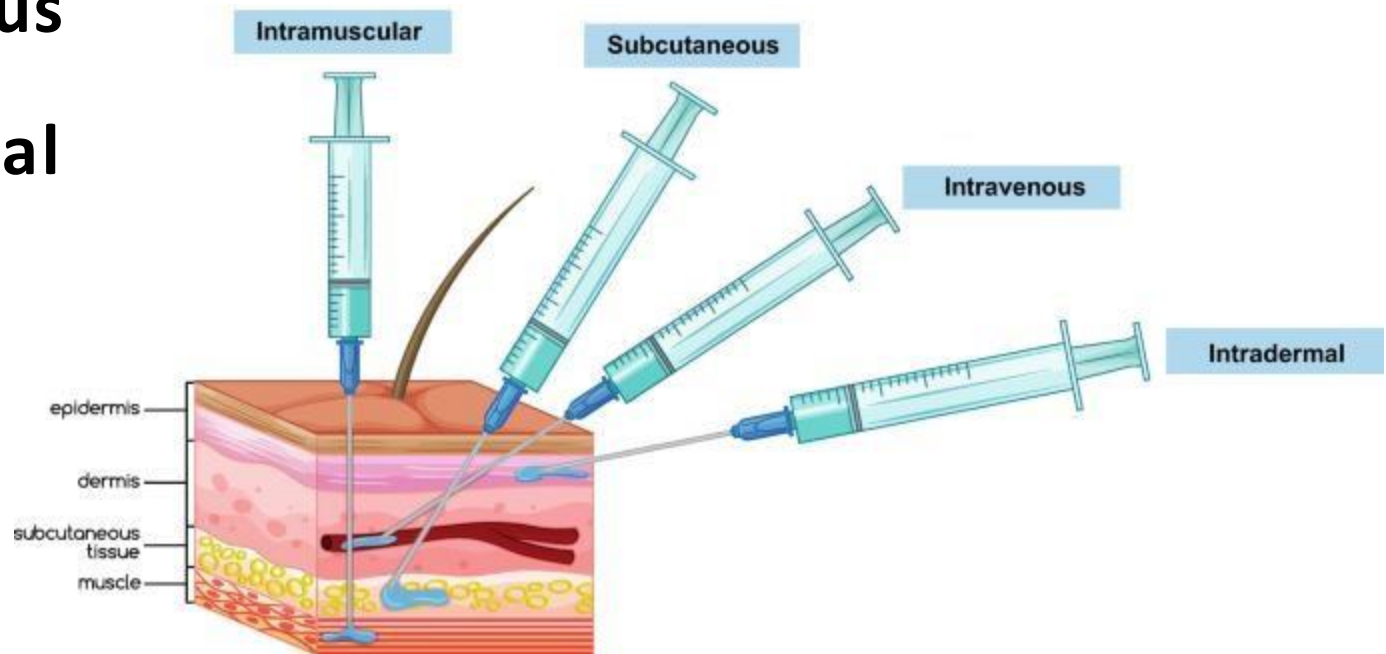
ENTERAL ROUTES

- **Sublingual:** the drug enters systemic circulation directly. The drug bypasses first-pass effect.
- **Rectal:** partial avoidance of the first-pass effect.



PARENTERAL ROUTES

- Intramuscular
- Subcutaneous
- Intravenous
- Intradermal



Tubic-Grozdanis, M., Krämer, I. (2023). Parenteral. In: Le Brun, P., Crauste-Manciet, S., Krämer, I., Smith, J., Woerdenbag, H. (eds) Practical Pharmaceutics. Springer, Cham. https://doi.org/10.1007/978-3-031-20298-8_21

OTHER ROUTES

- Oral Inhalation



- Nasal Inhalation



- Topical



- Transdermal



FACTORS THAT INFLUENCE DRUG ABSORPTION

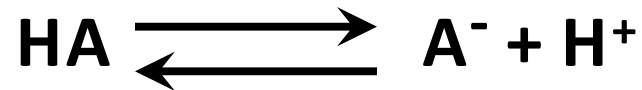
- Effect of pH on drug absorption
- Surface area available for absorption
- Blood flow to absorption site
- Contact time at absorption surface
- P-Glycoprotein

EFFECT OF pH ON DRUG ABSORPTION

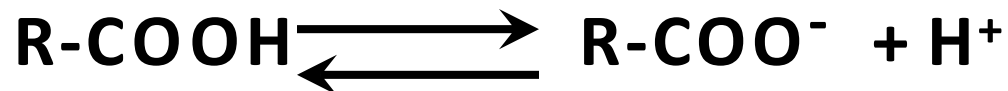
- Most drugs are weak acids or weak bases that are present in solution as both the nonionized and ionized species.
- The nonionized molecules are usually liposoluble and can diffuse through the cell membrane.
- The ionized molecules are usually unable to penetrate the lipid membrane because of their low liposolubility.

IONIZATION OF WEAK ACIDS

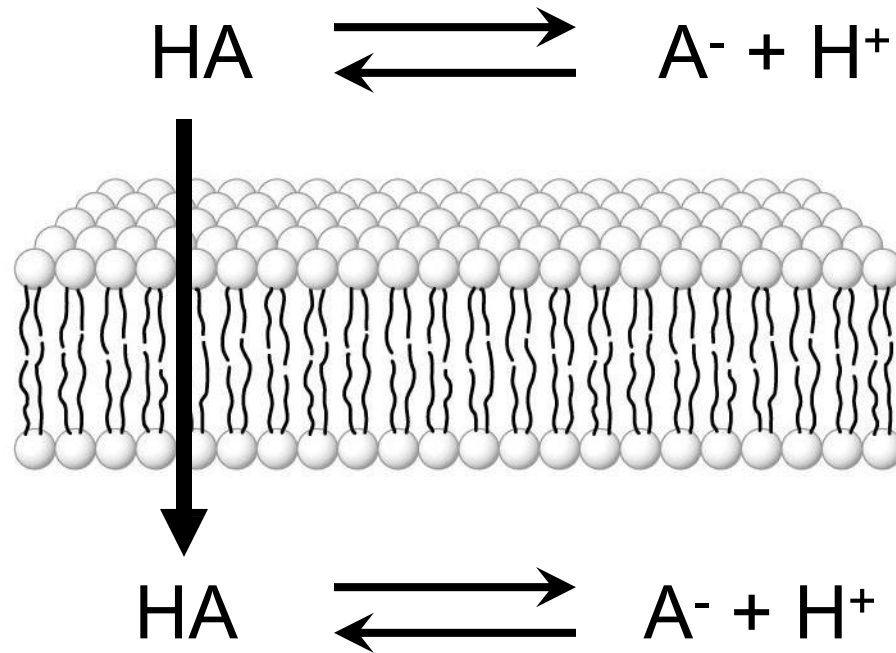
- For a **weak acid** the ionization reaction is:



- The uncharged form HA (ie the protonated form) will permeate through membranes.
- The charged form A⁻ (ie the unprotonated form) will not.
- Example:

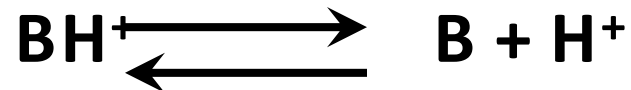


DIFFUSION OF WEAK ACIDS

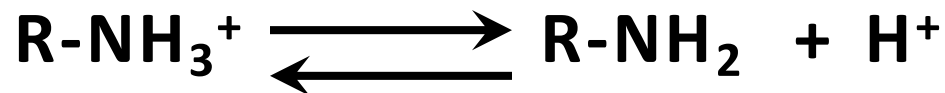


IONIZATION OF WEAK BASES

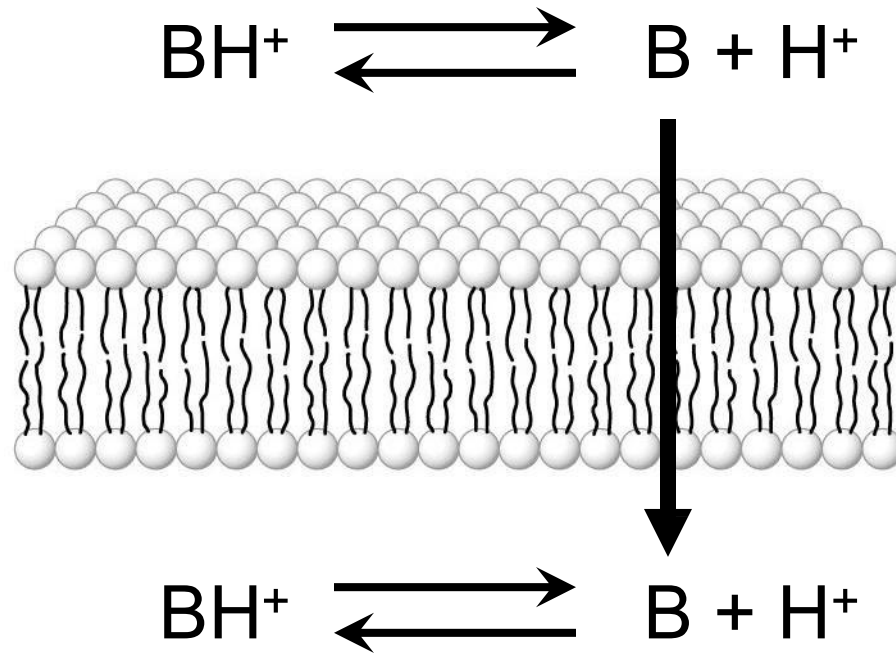
- For a **weak base** the ionisation reaction is:



- The uncharged form B (ie the unprotonated form) will penetrate through the cell membrane.
- The charged form BH⁺ (ie the protonated form) will not.
- Example:



DIFFUSION OF WEAK BASES



EFFECT OF pH ON DRUG ABSORPTION

In summary:

- **The protonated form of a weak acid is the more liposoluble form**
- **The unprotonated form of a weak base is the more liposoluble form.**

THE HENDERSON-HASSELBALCH EQUATION

- The ratio between the protonated and unprotonated forms is a function of the pH and of the pK of the ionizable group.
- The Henderson-Hasselbalch equation states that:

$$\text{pH} - \text{pK} = \log \frac{[\text{Unprotonated form}]}{[\text{Protonated form}]}$$

MORE ON THE H-H EQUATION

$$\text{pH} - \text{pK} = \log \frac{[\text{Unprotonated form}]}{[\text{Protonated form}]}$$

- When $\text{pH} = \text{pK}$, $[\text{unprotonated}] = [\text{protonated}]$
- For each unit of pH above the pK, the $[\text{unprotonated}]$ will be ten times the $[\text{protonated}]$.
- For each unit of pH below the pK, the $[\text{protonated}]$ will be ten times the $[\text{unprotonated}]$.

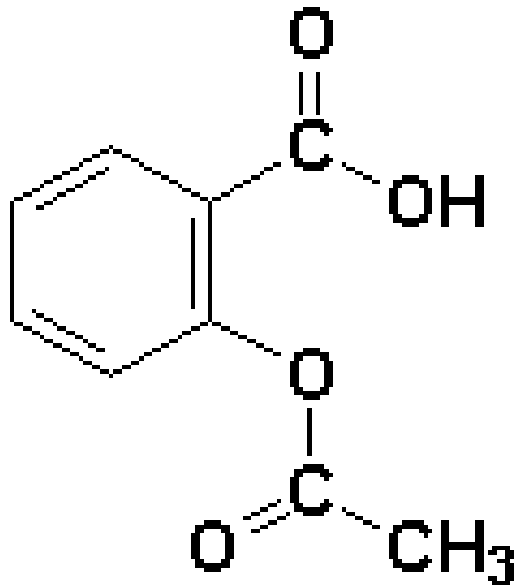
MORE ON THE H-H EQUATION

- We can construct a table with the % of the weak acid and the conjugate base as a function of the pH of the solution:

pH – pK	$\log \frac{[\text{unprotonated}]}{[\text{protonated}]}$	$\frac{[\text{unprotonated}]}{[\text{protonated}]}$	% unprotonated	% protonated
3	3	10^3	99.9	0.1
2	2	10^2	99	1
1	1	10	90	10
0	0	1	50	50
-1	-1	10^{-1}	10	90
-2	-2	10^{-2}	1	99
-3	-3	10^{-3}	0.1	99.9

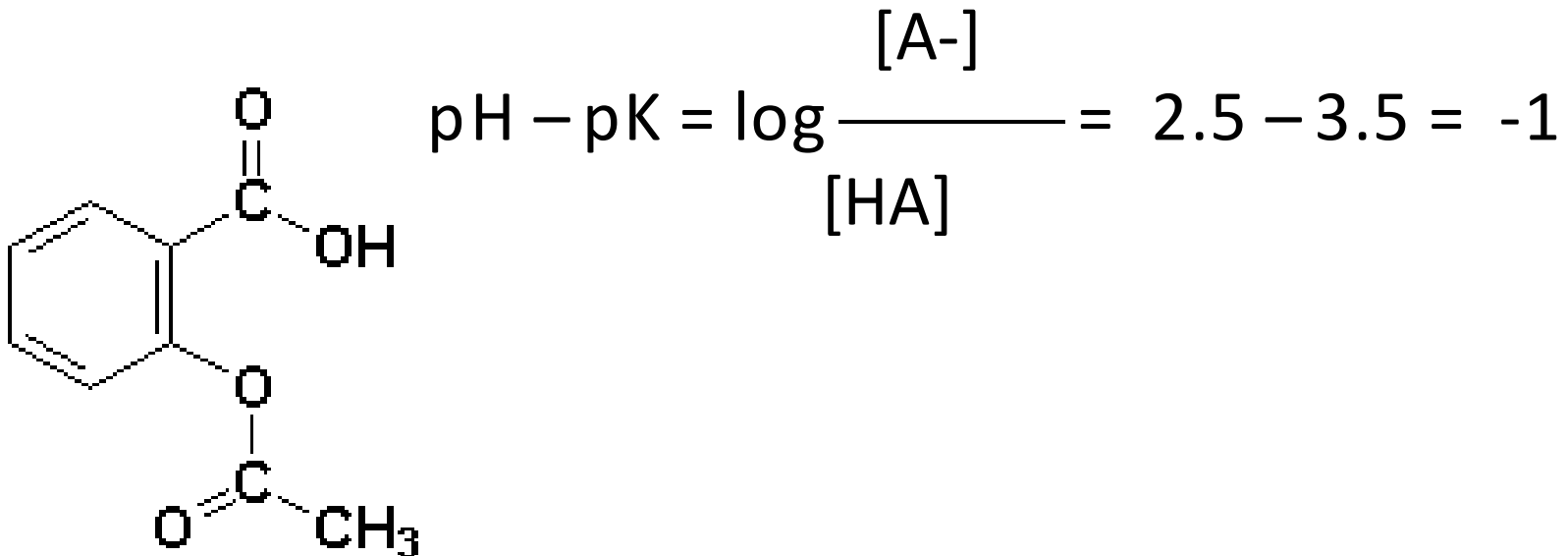
EXAMPLE

- Aspirin is a weak organic acid with a $pK = 3.5$.
What percentage of a given dose of aspirin will be in the liposoluble form at a stomach pH of 2.5?



EXAMPLE

- **ANSWER:** Aspirin is an acid, so it will be more ionized at alkaline pH and less ionized at acidic pH. From the Henderson-Hasselbalch equation we know that:



Therefore:

$$\frac{[A^-]}{[HA]} = 0.1$$

Therefore:

$$[A^-] = 10\%$$

$$[HA] = 90\%$$

EXAMPLE

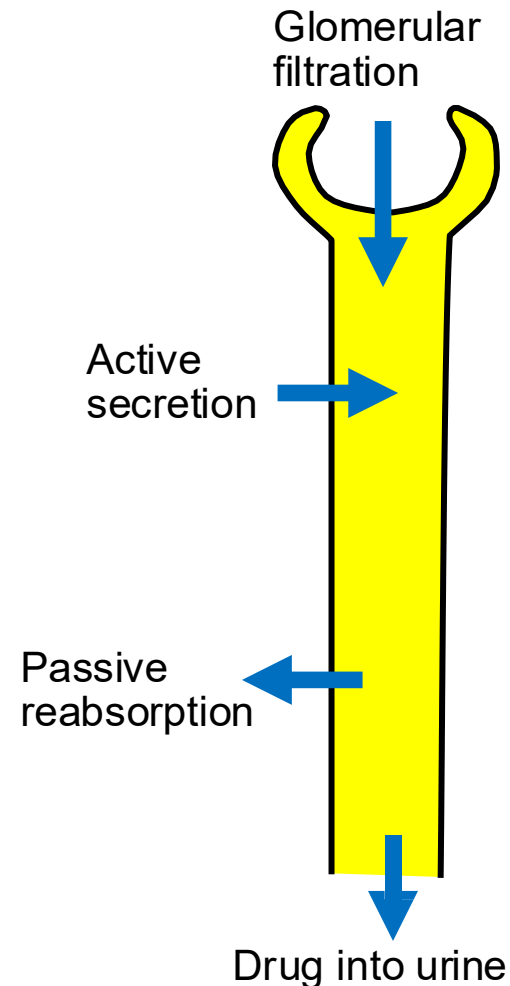
- For weak acids, like aspirin, the more **liposoluble form** is the protonated (uncharged) form: HA.
- Therefore, the % of a given dose of aspirin in the **liposoluble** form at a stomach pH of 2.5 is, then, **about 60%.**

pH – pK	$\log \frac{[\text{unprotonated}]}{[\text{protonated}]}$	$\frac{[\text{unprotonated}]}{[\text{protonated}]}$	% unprotonated	% protonated
3	3	10^3	99.9	0.1
2	2	10^2	99	1
1	1	10	90	10
0	0	1	50	50
-1	-1	10^{-1}	10	90
-2	-2	10^{-2}	1	99
-3	-3	10^{-3}	0.1	99.9

APPLICATIONS OF THE H-H EQUATION

ION TRAPPING

- The most important application of this equation is in the **manipulation of drug excretion by the kidney.**
- If a drug is in a liposoluble form during its passage down the renal tubule, a significant fraction will be reabsorbed by passive diffusion.



APPLICATIONS OF THE H-H EQUATION

- If the goal is to accelerate excretion of the drug, it is important to prevent its reabsorption from the tubule.
- This can be accomplished by adjusting urine pH to ionize the drug.
- As a result the drug will be “trapped” in the urine.
- **Weak acids are excreted faster in alkaline urine; weak bases are excreted faster in acidic urine.**

SURFACE AREA AVAILABLE FOR ABSORPTION

- **Most drug absorption occurs at the small intestine because of its large surface area.**

BLOOD FLOW TO THE ABSORPTION SITE

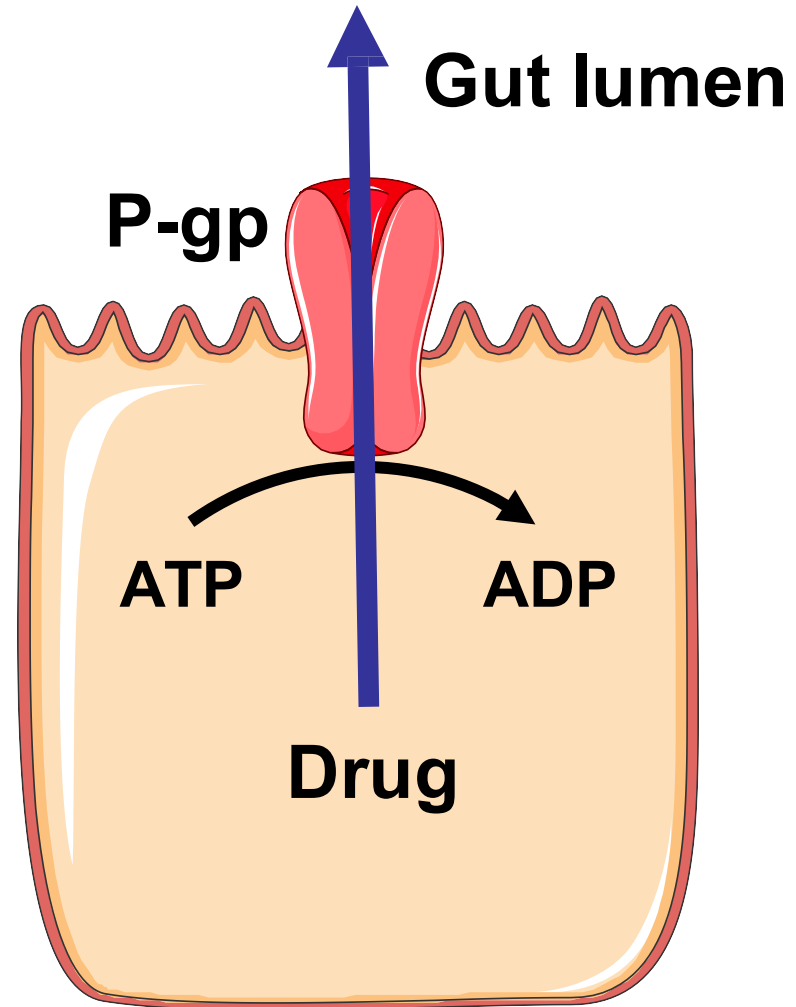
- **Blood flow to the intestine is much greater than the flow to the stomach:** absorption from the intestine is favored over that from the stomach.

CONTACT TIME AT THE ABSORPTION SURFACE

- If a drug moves through the GI tract very quickly, as in severe diarrhea, it is not well absorbed.

P-GLYCOPROTEIN

- P-glycoprotein (MDR1) is a transporter protein responsible for transporting several drugs across cell membranes.
- P-glycoprotein reduces drug absorption.



BIOAVAILABILITY (F)

- Fraction of administered dose of a drug that reaches the systemic circulation.

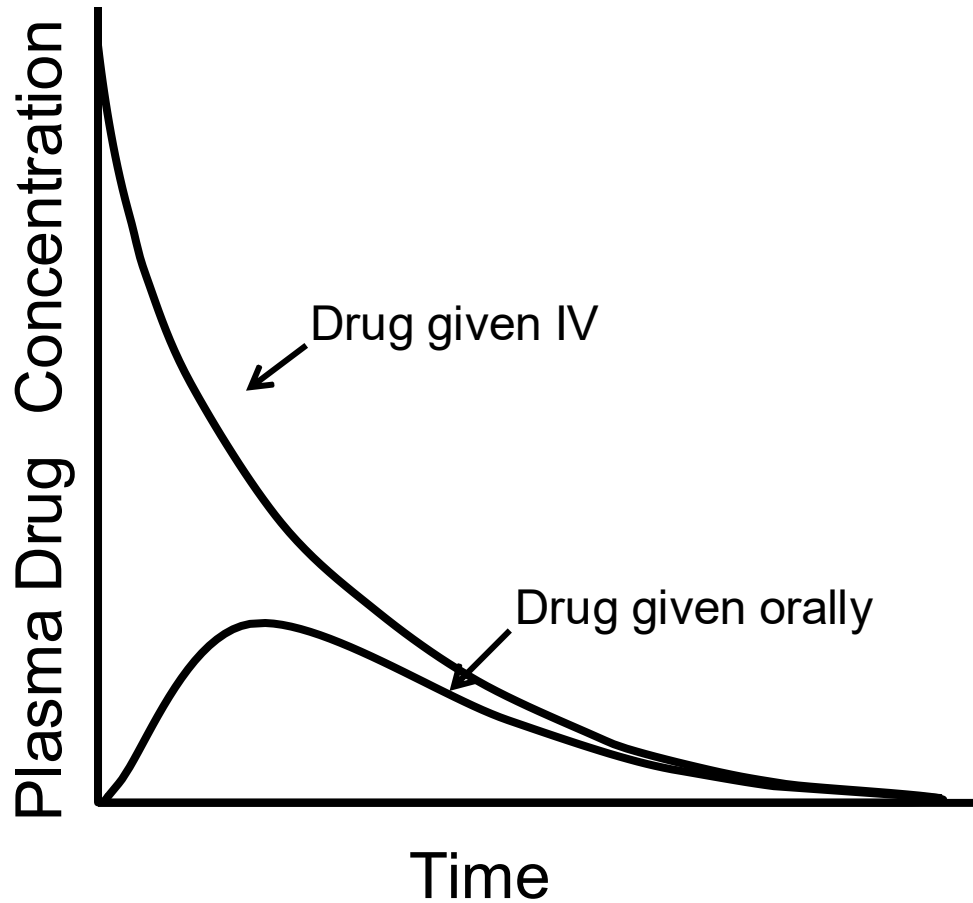
DETERMINATION OF BIOAVAILABILITY

- By plotting plasma concentrations of the drug vs. time, the area under the curve (AUC) can be measured.
- Bioavailability is determined comparing the AUC after a particular route of administration with the AUC after IV injection.

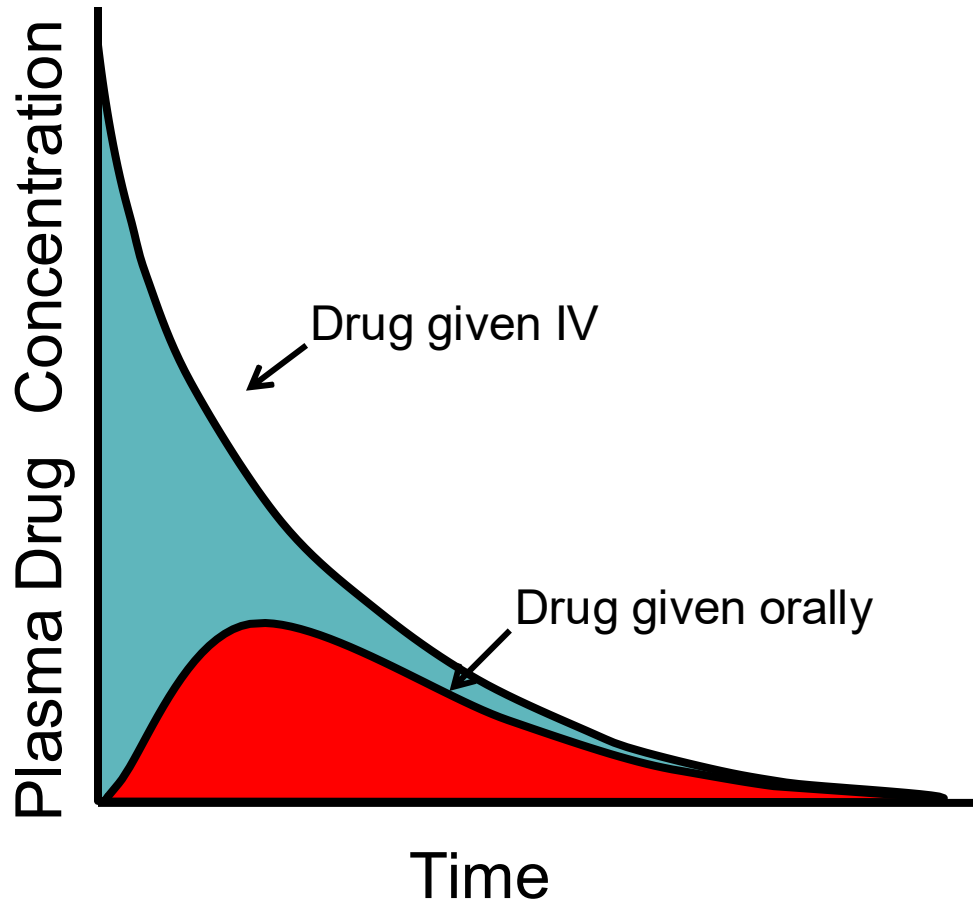
DETERMINATION OF BIOAVAILABILITY

- The AUC reflects the **extent of absorption** of the drug.
- For drugs administered IV, bioavailability is 100% by definition.

DETERMINATION OF BIOAVAILABILITY



DETERMINATION OF BIOAVAILABILITY



- Bioavailability is calculated as:

$$F = \frac{\text{AUC oral}}{\text{AUC IV}} \times 100$$

DRUG DISTRIBUTION

- It is the process by which a drug leaves the blood stream and enters the extracellular fluid and /or the cells of the tissues.

FACTORS INFLUENCING DRUG DISTRIBUTION

- The amount of drug distributed into tissues is mainly determined by:
 - Blood flow
 - Drug binding to plasma proteins
 - Drug tissue binding
 - Drug liposolubility

BLOOD FLOW

- Initially, liver, kidney, brain, and other well-perfused organs receive most of the drug.
- Delivery to muscle, most viscera, skin, and fat is slower.

DRUG BINDING TO PLASMA PROTEINS

- Many drugs circulate in the bloodstream bound to plasma proteins.
- Reversible binding to plasma proteins slows distribution from the blood to other tissues.

DRUG TISSUE BINDING

- Many drugs accumulate in tissues.
- Many liposoluble drugs accumulate in fat.
- Accumulated drug may serve as a reservoir that prolongs drug action.

DRUG LIPOSOLUBILITY

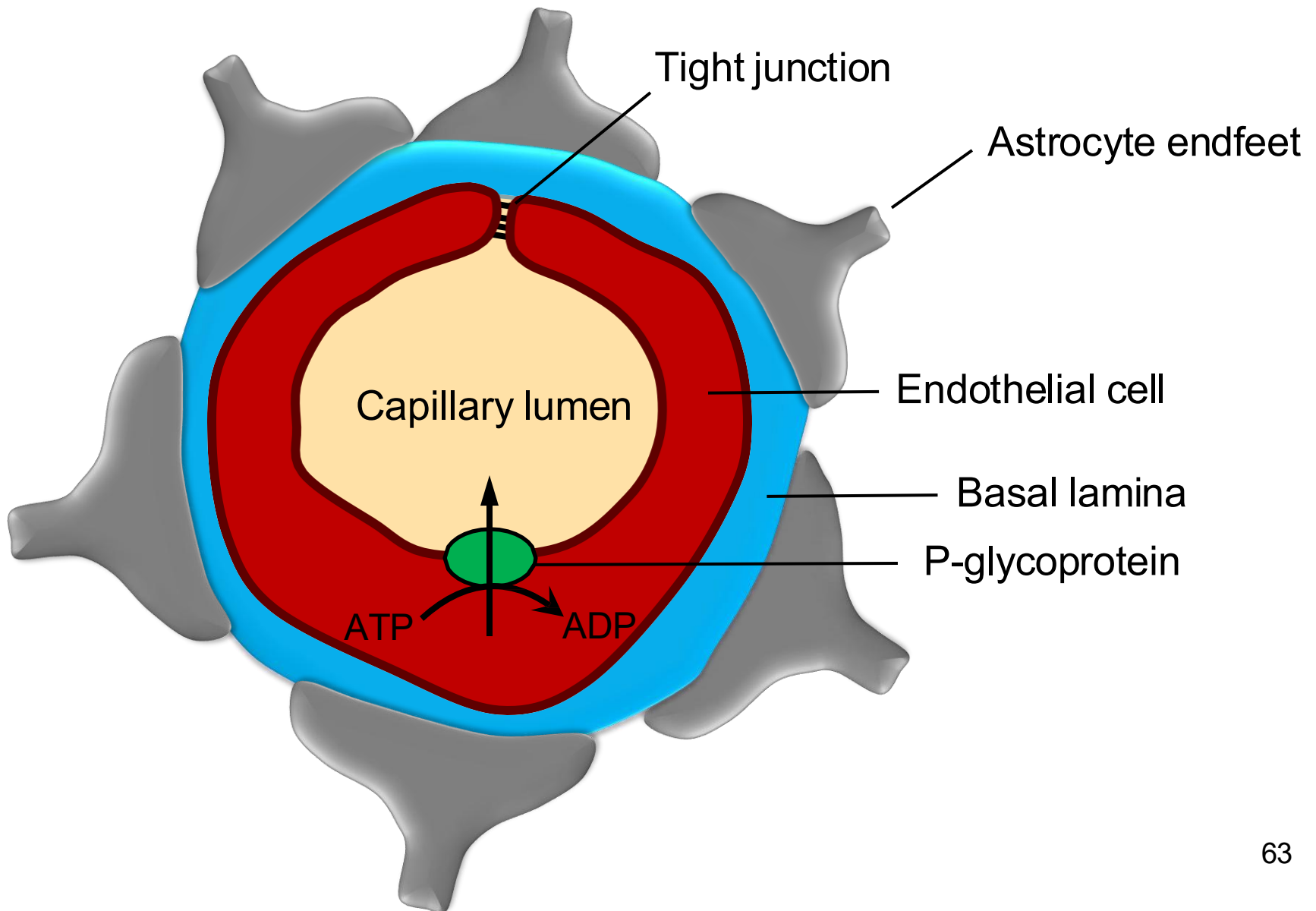
- Drug liposolubility is a major factor affecting drug distribution, particularly to the brain, where the **blood-brain barrier** restricts the penetration of polar molecules.
- Drug penetration into the brain depends on transcellular rather than paracellular transport.

DRUG LIPOSOLUBILITY

THE BLOOD-BRAIN BARRIER

- The endothelial cells of the brain capillaries have continuous tight junctions.
- Astrocytic “end feet” surround the outside of capillary endothelial cells.
- The P-glycoprotein actively transports drugs back into the systemic circulation.

THE BLOOD-BRAIN BARRIER



THE BLOOD-BRAIN BARRIER

- To enter the CNS drugs must be **highly liposoluble or have an active transporter.**

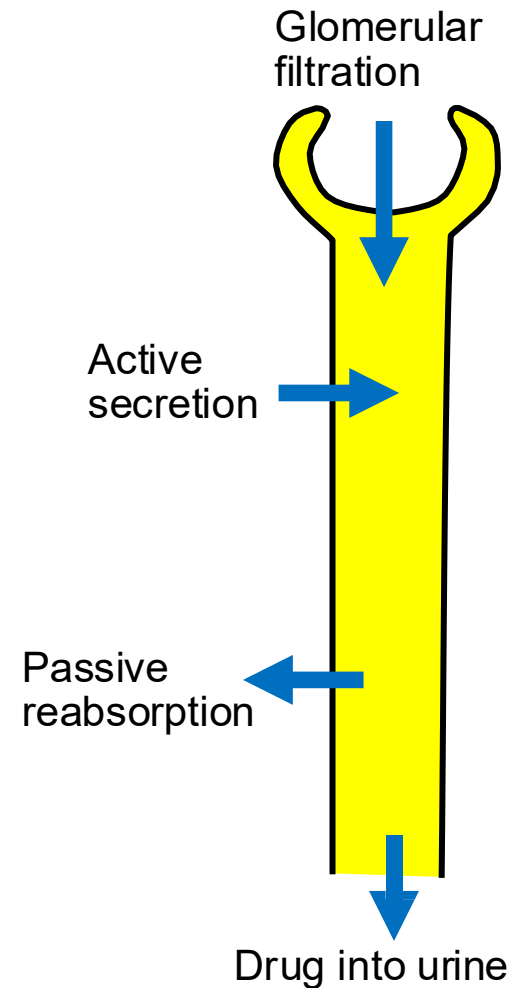
DRUG ELIMINATION

- It occurs by two processes:
 - **Metabolism**
 - **Excretion**

METABOLISM OF DRUGS:

WHY IS DRUG METABOLISM NECESSARY?

- **Hydrophilic drugs** are typically directly excreted by the kidneys.
- But most drugs are **lipophilic**.
- This property enables drugs to pass across cell membranes and gain access to their site of action.
- Lipophilic compounds are not easily excreted by the kidney because they are **reabsorbed through the tubular membranes**.
- Therefore, lipophilic drugs are first metabolized into more hydrophilic metabolites.



METABOLISM OF DRUGS:

WHY IS DRUG METABOLISM NECESSARY?

- Metabolism of drugs into more hydrophilic metabolites is essential for
 - **Termination of their biological activity**
 - **Their elimination from the body.**

METABOLISM OF DRUGS

- In general, metabolism generates **more polar, inactive metabolites** that are readily excreted from the body.
- However, in some cases, metabolites with biological activity or even toxic properties are generated.

REACTIONS OF DRUG METABOLISM

- Drug metabolism involves two kinds of biochemical reactions: **phase I and phase II**.
- These often occur sequentially.

PHASE I REACTIONS

- Phase I reactions include **oxidations, reductions, decarboxylations, deaminations and hydrolytic reactions.**
- Phase I reactions usually convert the parent drug to a more polar metabolite.
- Phase I reactions generally result in inactive metabolites.

PHASE II REACTIONS

- **Phase II consists of conjugation reactions.**
- **Phase II** conjugation reactions form a covalent bond between the drug molecule and **glucuronate, acetate, glutathione, amino acids or sulfate.**

REACTIONS OF DRUG METABOLISM

ABSORPTION

METABOLISM

ELIMINATION

Phase I

Phase II

Drug → Metabolite → Conjugate →

Drug → Metabolite →

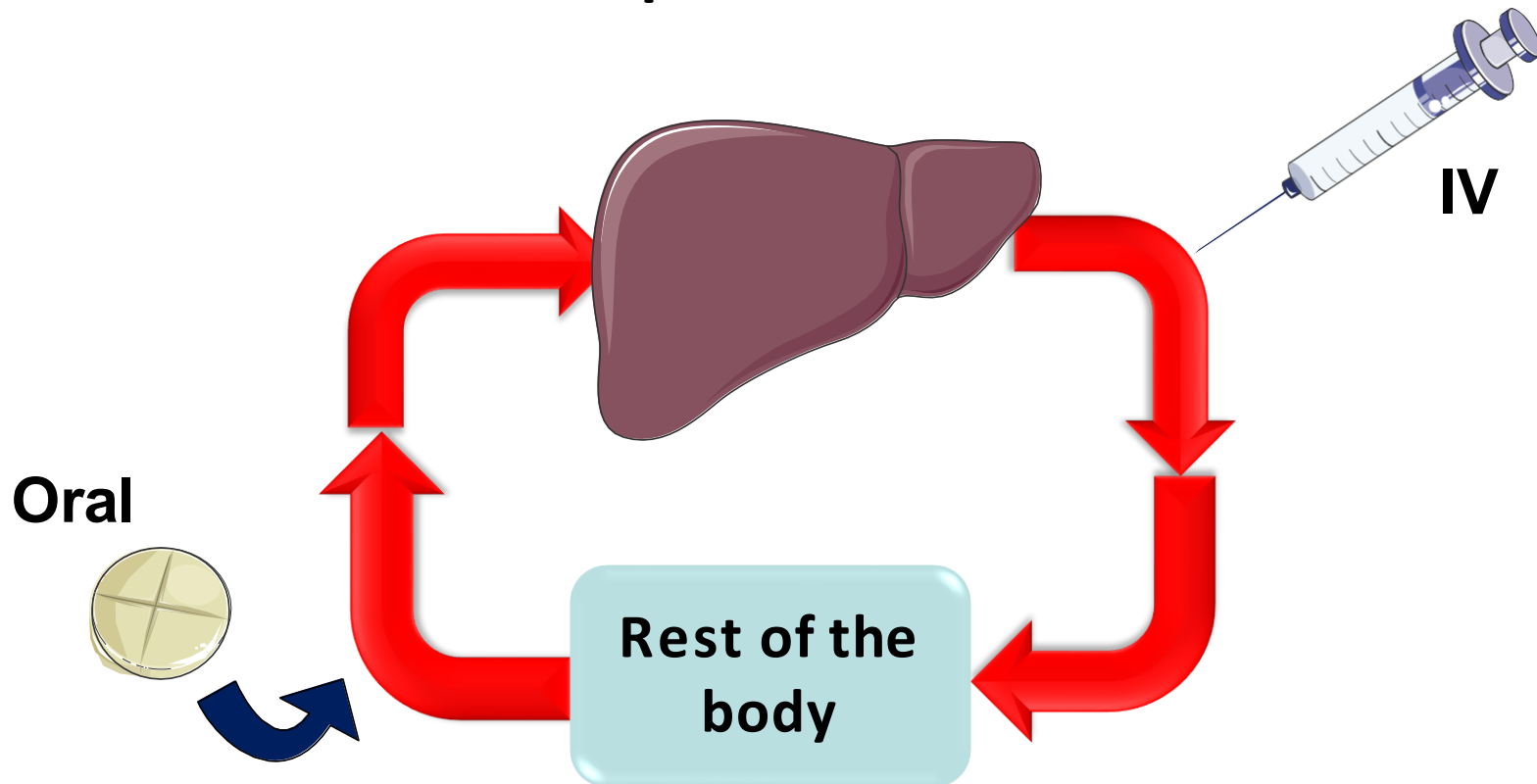
Drug → Conjugate →

Drug →

Hydrophilic → **Hydrophilic**

SITES OF DRUG METABOLISM

- The **liver** is the main organ of drug metabolism.
- This fact figures prominently in the phenomenon known as the **first-pass effect**.



SITES OF DRUG METABOLISM

- The liver is quantitatively the most important organ in metabolizing drugs.
- But every tissue in the body is capable of drug metabolism to some degree.
- Particularly active sites include the **the GI tract, the kidneys, the skin, and the lungs.**
- The GI tract also contributes to the first-pass effect.

CYTOCHROME P450

- Most phase I reactions are catalyzed by the cytochrome P450 system.
- The cytochrome P450 is a superfamily of heme-containing isozymes.
- Enzymes of the **cytochrome P450 system**, are responsible for the metabolism of 80% of all drugs in clinical use.

CYTOCHROME P450

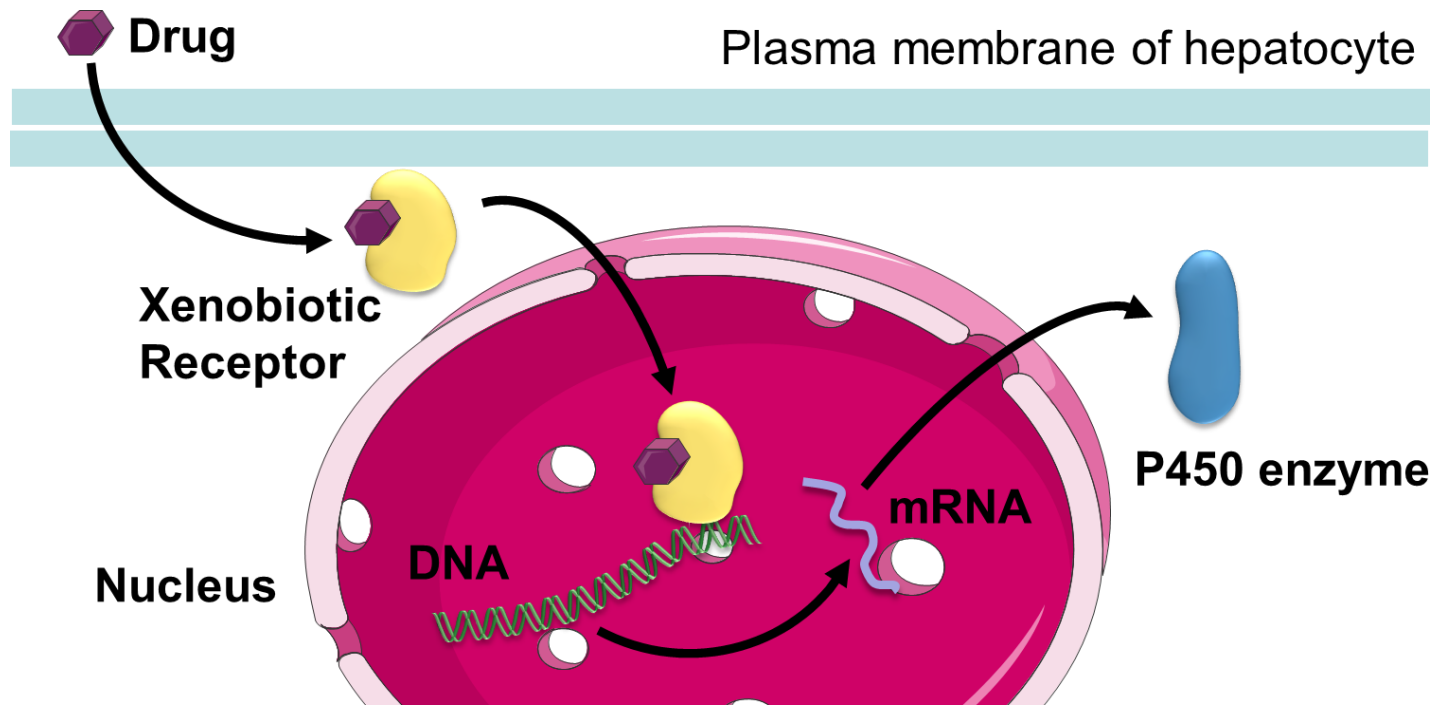
- The P450 proteins are categorized into families and subfamilies by their sequence similarities.
- Humans have 18 families of cytochrome P450 genes divided into 43 subfamilies.
- **Families 1-3 are mostly active in drug metabolism.**
- The two most important P450 isozymes in drug metabolism are CYP3A4 and CYP2D6.

ENZYME INDUCTION

- Some cytochromes P450 are inducible.
- P450 enzyme induction typically occurs via **increased transcription.**
- Certain drugs increase the synthesis of one or more P450 isoforms.
- Examples are:
 - **Rifampin**
 - **Phenobarbital**
 - **Carbamazepine**
 - **St John's wort**

ENZYME INDUCTION: MOLECULAR MECHANISM

- Drugs can enter hepatocytes and bind to **xenobiotic nuclear receptors**.
- The drug-receptor complex translocates to the nucleus and activates the transcription of genes coding for P450 isozymes.

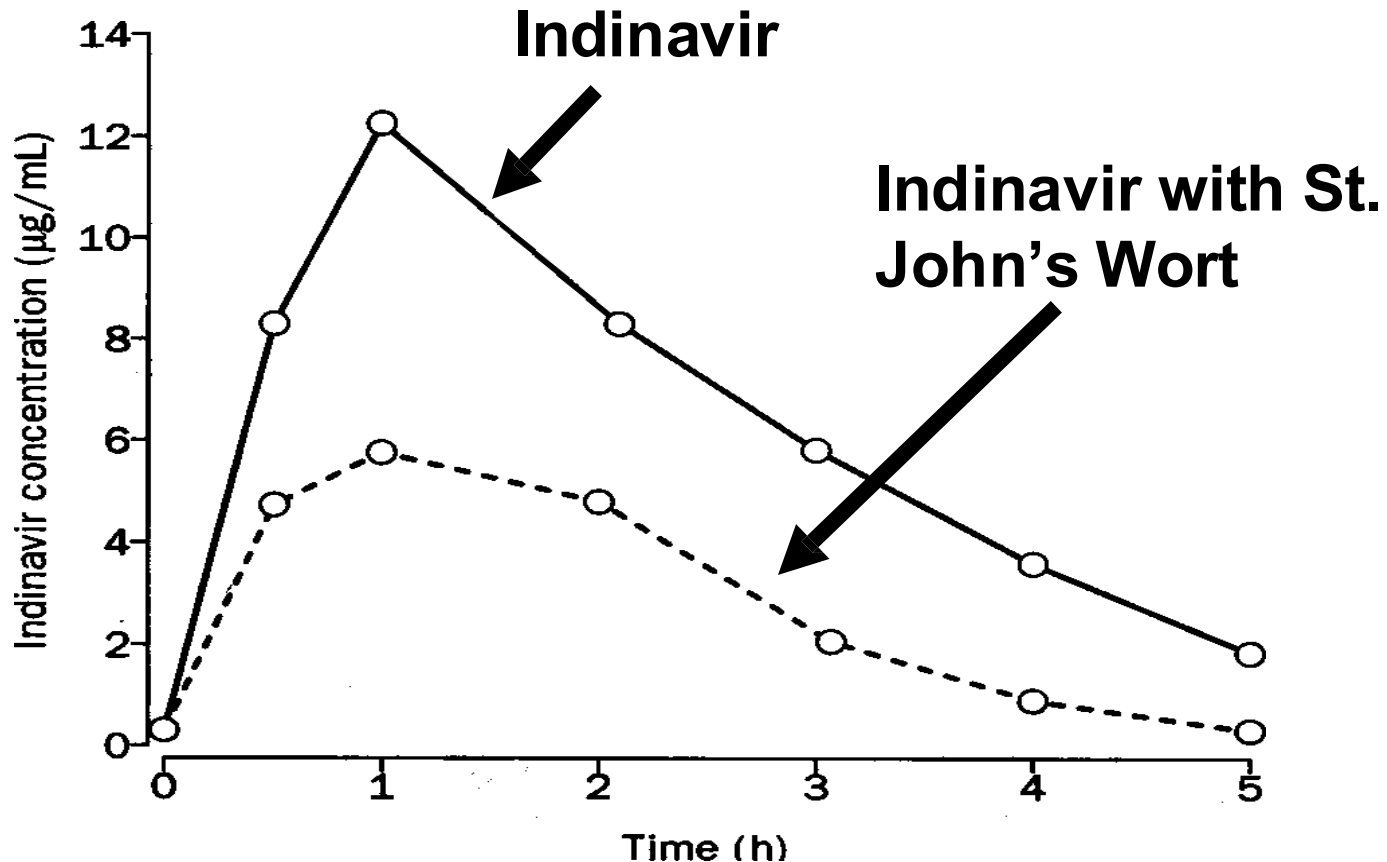


ENZYME INDUCTION: CLINICAL CONSEQUENCES

- Induction has important clinical consequences.
 - A drug can increase its own metabolism.
 - A drug can increase the metabolism of a coadministered drug.
 - **This may reduce drug plasma concentrations below therapeutic levels.**

ENZYME INDUCTION: CLINICAL CONSEQUENCES

- St. John's wort contains hyperforin.
- Hyperforin increases the metabolism of various drugs, by inducing CYP3A4.



ENZYME INHIBITION

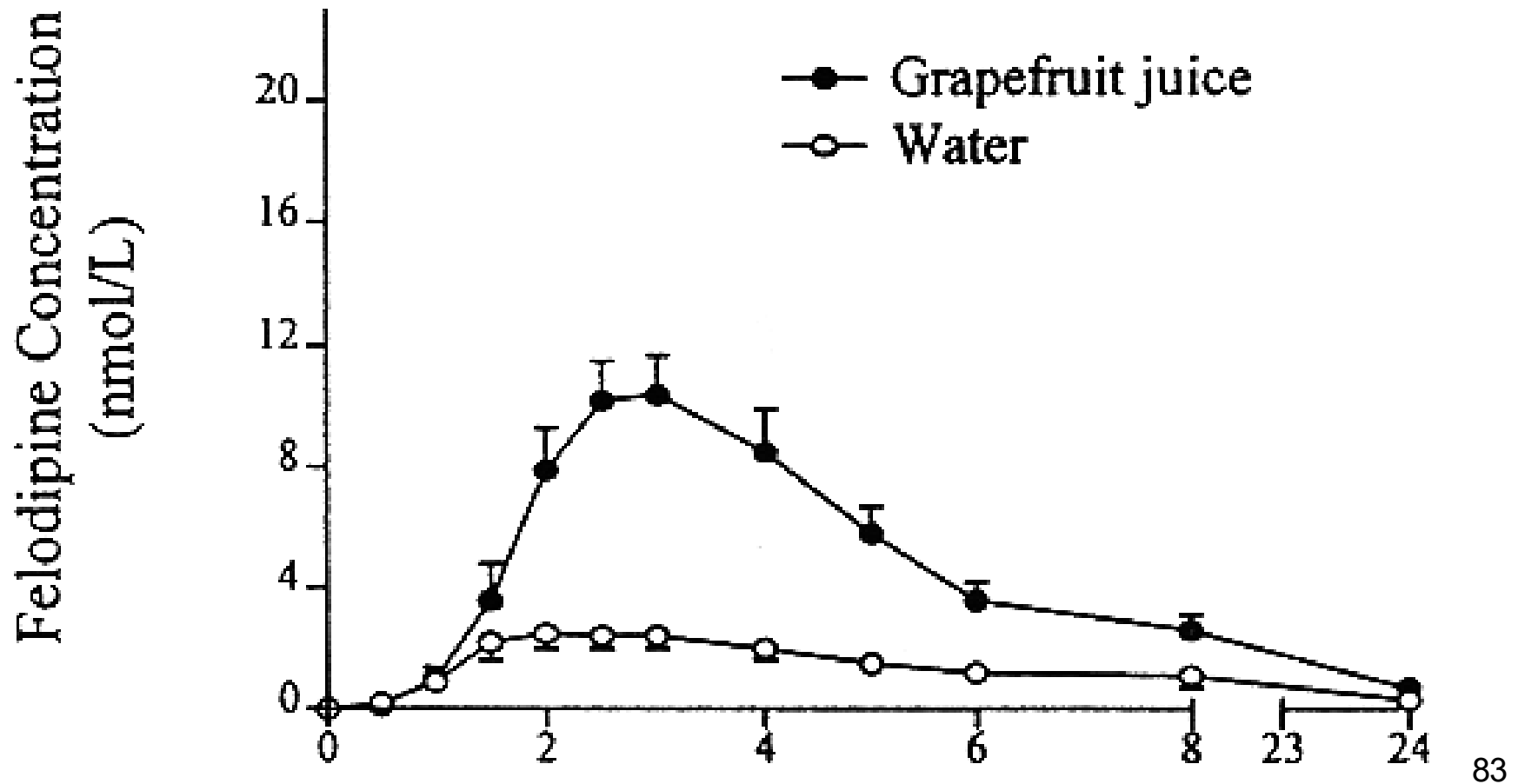
- Certain drugs inhibit cytochromes P450.
- Typical examples are:
 - **Cimetidine**
 - **Erythromycin**
 - **Chloramphenicol**
 - **Grapefruit juice**

ENZYME INHIBITION: CLINICAL CONSEQUENCES

- The metabolism of drugs that are metabolized by the inhibited enzyme will decrease.
- Drug levels may reach **toxic concentrations**.

ENZYME INHIBITION: CLINICAL CONSEQUENCES

- Grapefruit juice inhibits felodipine metabolism.



DRUG EXCRETION

- The reactions of metabolism enhance the hydrophilicity of drugs and their metabolites, enabling them to be excreted.
- **Renal excretion** is the most common mechanism of drug excretion.
- A small number of drugs are excreted in the **bile**.



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